

HOW TO PLAY — Snake Battle

Two-player snake on one keyboard, written in pure Python `turtle`. **The rules are not normal snake rules: biting punishes the BITER.** Read section 4 before your first match.

1. Start

```
python snake_battle.py
```

Needs Python 3.8+ with `tkinter` (the python.org installer for Windows and macOS already includes it · Ubuntu: `sudo apt install python3-tk`). **Nothing else to install.**

From the menu pick **1 PLAYER** or **2 PLAYER BATTLE** → choose a character and a power → `Enter`.

2. Controls

Character Select

	Change character	Change power
P1	<code>A / D</code>	<code>W / S</code>
P2	<code>← / →</code>	<code>↑ / ↓</code>

You can also click the ◀▶ arrows or click a power icon · `Enter` starts · `Esc` goes back.

Two players cannot take the same character. The arrows skip whatever the other player is holding. Powers may be duplicated.

In game

	1 Player	2 Player
Move	<code>↑ ↓ ← →</code>	P1 <code>W A S D</code> · P2 <code>↑ ↓ ← →</code>
Use power	<code>Space</code>	P1 <code>Q</code> · P2 <code>O</code>
Sound on / off	<code>X</code>	<code>X</code>
Restart / next round	<code>R</code>	<code>R</code> (on the result screen)
Back to menu	<code>M</code>	<code>M</code>

3. Goal

Win a round by either:

- dropping the other player's HP to **0** (both start with 3 hearts), or
- reaching **800 points** first (a fruit is 50 → about 16 fruit)

Win the match by taking **2 rounds** (best of 3).

4. The core rule — read this one to the end

Normal snake: touch anything and you die. Not here.

4.1 Biting someone costs the BITER

Put your head into the other snake's body or tail → **you lose 1 HP**, not them. No points change hands, even if you are the one behind on score.

What follows from that: the player who is ahead wants to *dangle* their body and tail in front of the other player's mouth, and the player who is behind has to **resist biting** the bait.

4.2 Head-to-head — the LOWER score wins, and gets paid

Situation	Result
Scores differ	The player ahead loses 1 HP and hands over half of their own score · the player behind loses nothing
Scores exactly level	Both lose 1 HP · no transfer

This is the second way to score, next to eating fruit — if you are behind, ramming head-first is profit.

Remember: this is the only place points change hands. Every other collision costs HP alone.

4.3 Everything at once

What you did	HP	Score
Head clash, you were ahead	-1	half of it transfers to the other player
Head clash, you were behind	—	you gain points
Bit the other snake's body / tail	-1	—
Got bitten	—	—
Hit your own body	-1	—
Hit a wall or a box	—	— (stunned for 0.5 s)

After losing HP you blink and are **immune for 1 second**, so nothing can chain-hit you.

5. Powers — one per player

The meter fills to 100 · **one fruit** = **+25** → a power roughly every 2 fruit.

Power	What it does	Cost	Duration
SPEED	move twice as fast	50	~3 s
STEALTH	body disappears, head goes ghost (you still collide with everything)	50	~3 s
SHIELD	take no damage at all	50	~2.5 s
PHASE	pass through boxes and your own body	50	~2.5 s
FEAST	fruit is worth double	40	~4 s

STEALTH does not let you pass through anything — it only hides your body. The one that really passes through is PHASE.

A power cannot be re-triggered while it is still running.

6. Reading the screen

What you see	Where	Meaning
♥♥♥ + name + number	top left / top right	HP and score for this round
coloured bar	under the name	power meter (full at 100)
icon beside the bar	beside the bar	the power you picked — Q / 0 fires this
P1 1 - 0 P2	centre, gold	rounds won in this match
STUN / BOOST / CLOAK	floating over the snake	that snake's current state

STUN = frozen for 0.5 s (from a wall, a box, or losing HP).

7. Result screens

End of a round → **ROUND OVER** — who took the round, the round score, what is left to play for. R goes to the next round; the tally and the ledger carry over.

End of the match → **MATCH RESULT** — the whole match:

```

ROUND  P1 score x hp + bonus = MVP  P2 score x hp + bonus = MVP  WON BY
R1      400 x 3 + 300 = 1500          200 x 0 + 0 = 0          SIAM
R2      150 x 0 + 0 = 0              480 x 2 + 200 = 1160        SAKURA
R3      300 x 0 + 0 = 0              300 x 0 + 0 = 0          DRAW
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MVP TOTAL          1500                      1160          MVP: P1

```

MVP = (score + 100) × HP left, added up over every round.

- **Every heart you finish a round with is worth 100 points on its own.**
- **A round you were knocked out of (HP = 0) is worth nothing**, however many points you banked — everything is multiplied by 0.
- **MVP does not decide the match.** Rounds won decide it. MVP is only the tie-breaker when the rounds are level (1-1 after three rounds, say). Level on MVP too is a drawn match.

R here starts a fresh match at 0-0 · M abandons it and goes back to the menu.

8. Tips

If you are ahead

- Avoid head clashes — you lose every one of them and hand over half your score.
- Park your body and tail across the other player's mouth and let them bite.
- Being ahead already walks you toward the 800-point win. Do not gamble.
- SHIELD is worth most when ahead: it blocks the HP loss **and** the score transfer.

If you are behind

- Ram head-first. You win the clash and get points for free.
- Do not bite their body, however tempting — you are the only one who gets hurt.
- SPEED helps you catch their head.

General

- You grow one segment per fruit — the longer you are, the easier it is to hit yourself on a tight turn.
 - Walls and boxes cost no HP, only a stun — they are cover, not death.
 - Dying to your own body still costs HP, and a round you are knocked out of scores 0 MVP.
 - `X` toggles sound at any time.
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9. Troubleshooting

Symptom	Fix
No sound at all	Read the <code>SOUND:</code> line the game prints at startup — it says why. Usually the <code>.wav</code> files are not 16-bit PCM.
<code>ModuleNotFoundError: tkinter</code>	Linux: <code>sudo apt install python3-tk</code>
The text looks odd	The game picks a font the machine actually has; the candidate list is at the top of <code>snake_battle.py</code> .
Want to change difficulty / colours / speed	Section 3 of <code>snake_battle.py</code> — see <code>docs/MANUAL.md</code> section 8.

Technical detail — file map, adding sound, tuning, adding a character — is in `docs/MANUAL.md`.